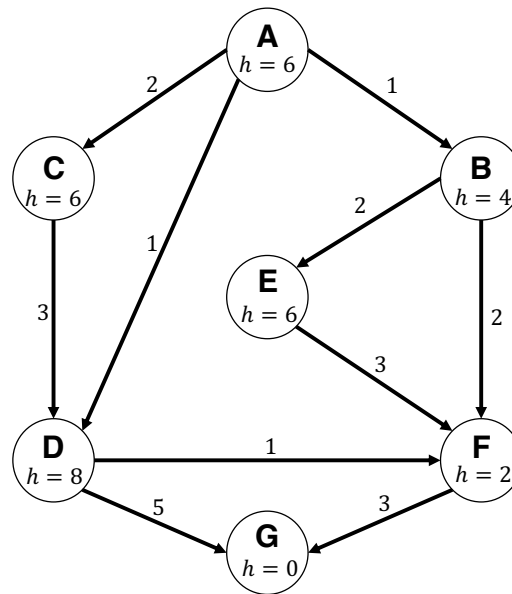


CS47100: Introduction to Artificial Intelligence

Midterm Exam

1. [15 points] Search

Consider the following state space graph, for which **A** is the **initial state**, **G** is the **goal state**, the cost for a step/action is given by the number next to each edge, and the value of the heuristic function $h(\cdot)$ is shown below the state (i.e., $h(A) = 6$, $h(B) = 4$, $h(C) = 6$, etc.).

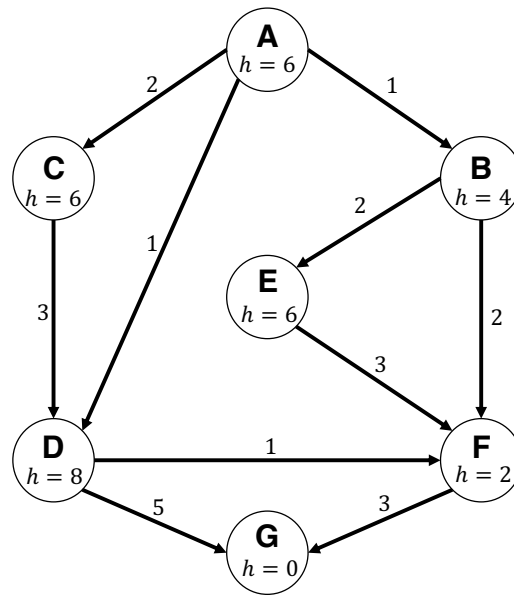


Recall that **best-first search** involves choosing the node from the frontier (implemented as a priority queue) with the **minimum** value of some evaluation function $f(n)$. For each of the following, use the **best-first search algorithm** with the **provided evaluation function** to plan a route from the initial state to the goal state, using the **tree search version** of the algorithm. Write down (i) the order of the nodes **expanded** when executing the **tree search version** of the algorithm, and (ii) the path that will actually be visited according to results from the search. Use **alphabetical order** for tie-breaking in the **priority queue**. In addition, be sure to check for a goal node only upon expanding the node.

For this problem, we will additionally let:

- $g(n)$ denote the path cost from the start node to the current node n
- $\text{Depth}(n)$ denote the depth of a node n
- $\text{Alpha}(n)$ denote the alphabetical index (starting from 1) of a given node (i.e. $\text{Alpha}(A) = 1$, $\text{Alpha}(B) = 2$, etc.)

Your Username:



- (a) [3 points] Evaluation function: $f(n) = \text{Depth}(n)$

Answer:

(1.5pts) order: A, B, C, D, D, E, F, F, G

(1.5pts) path: A, D, G

- (b) [3 points] Evaluation function: $f(n) = \text{Depth}(n) + \frac{1}{\text{Alpha}(n)}$

Answer:

(1.5pts) order: A, D, C, B, G

(1.5pts) path: A, D, G

- (c) [3 points] Evaluation function: $f(n) = \frac{1}{\text{Depth}(n)+1}$

Answer:

(1.5pts) order: A, B, E, F, G

(1.5pts) path: A, B, E, F, G

- (d) [3 points] Evaluation function: $f(n) = g(n) + h(n)$

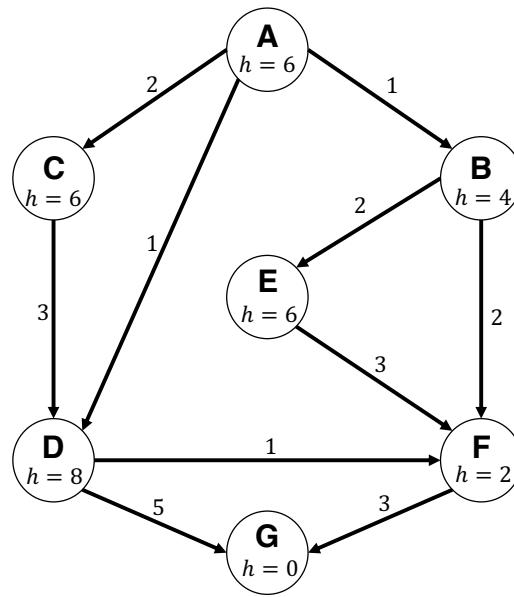
Answer:

(1.5pts) order: A, B, F, G

(1.5pts) path: A, B, F, G

- (e) [3 points] Evaluation function: $f(n) = g(n) + \frac{h(n)}{2}$

Your Username:



Answer:

(1.5pts) order: A, B, F, C, D, F, G

(1.5pts) path: A, D, F, G

2. [15 points] Games

In this problem, we model a game with max, min, and chance nodes. The upward-pointing triangle (max node) tries to maximize the outcome, the downward-pointing triangle (min node) tries to minimize the outcome, and the circle (chance node) chooses randomly, with the probability of each possible choice given by the value next to the edge pointing from a chance node to a min node. (In the game tree below, all such edges have probability $\frac{1}{2}$, meaning that the circle player chooses uniformly at random, i.e., its children nodes all have **equal chance** of being chosen.) The squares show the utility of a terminal node.

- (a) [4 points] What is the EXPECTED-MINIMAX-VALUE for each of the nodes MAX, E1, E2, MIN1, MIN2, MIN3, and MIN4? Show your work.

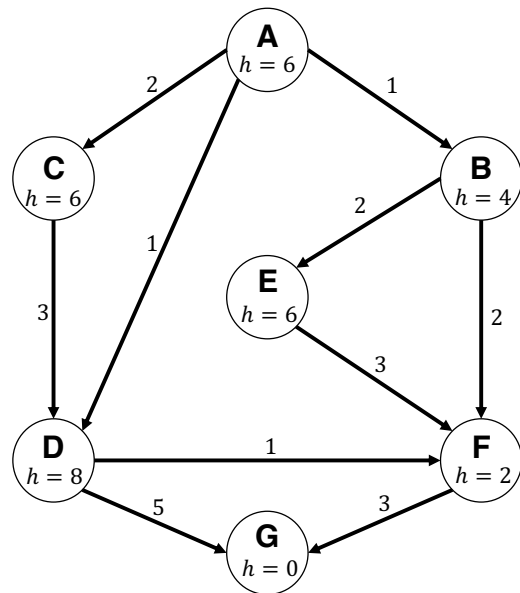
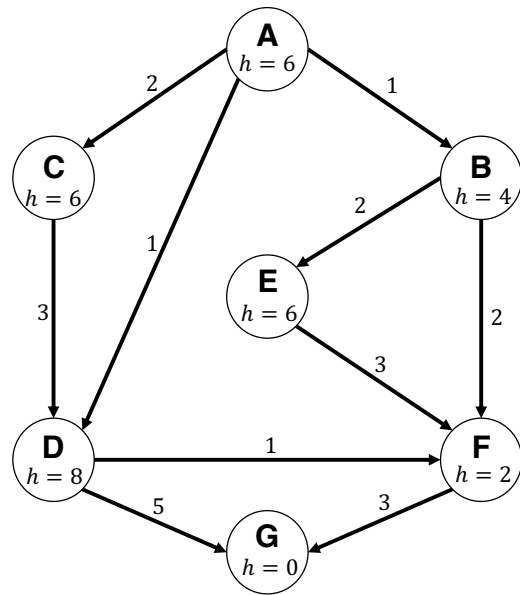
Answer: -0.5 for each incorrect EXPECTED-MINIMAX-VALUE.

MIN1 = $\min(4,1) = 1$, MIN2 = $\min(5,9) = 5$, MIN3 = $\min(2, 3) = 2$, MIN4 = $\min(6,7) = 6$,

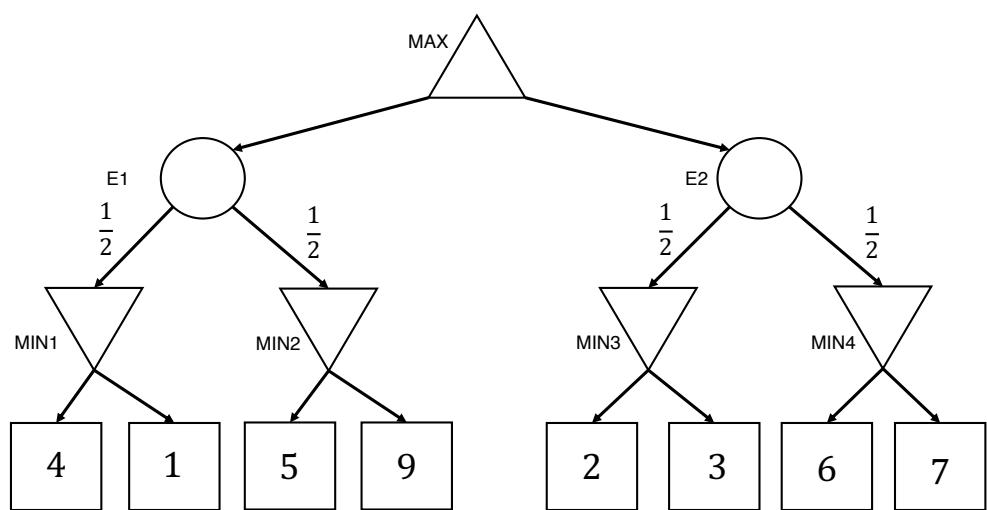
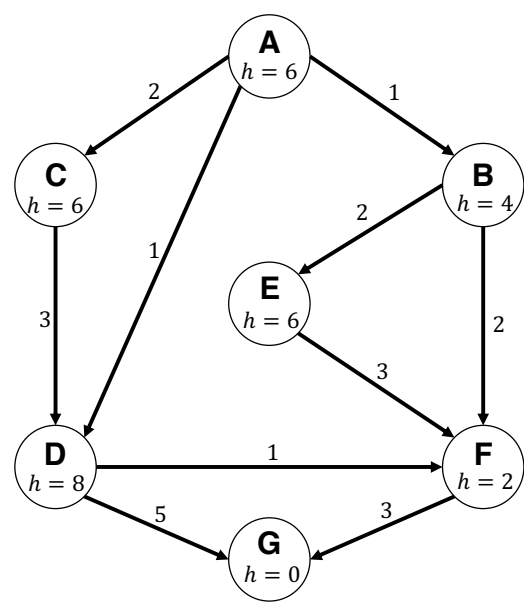
E1 = $(1+5)/2 = 3$, E2 = $(2+6)/2 = 4$,

MAX = $\max(3,4) = 4$.

Your Username:

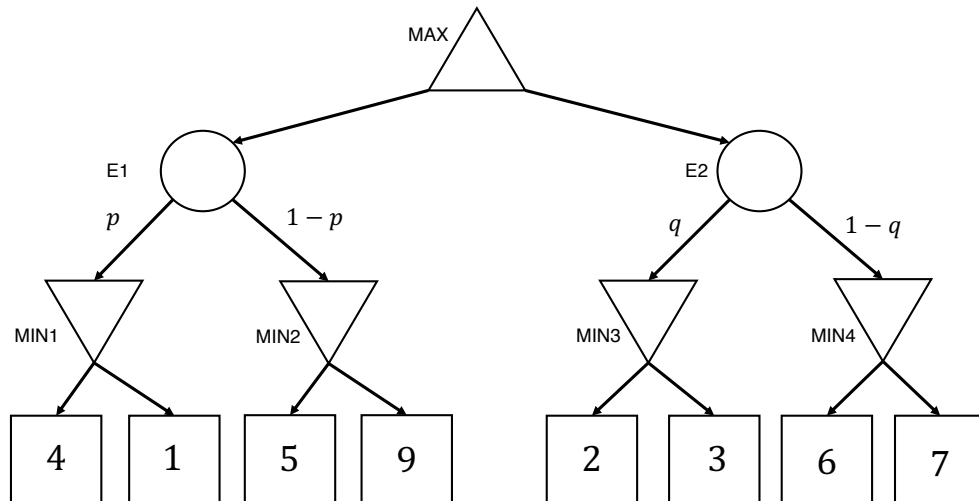


Your Username:



Your Username:

- (b) In the tree below, we now consider the case where the probabilities are expressed in terms of p and q , where p and q **are only allowed to be** such that $0 \leq p \leq 1$ and $0 \leq q \leq 1$.



- i. [3 points] Let $p = \frac{1}{2}$. Determine q such that the EXPECTED-MINIMAX-VALUE for E1 **is equal to** the EXPECTED-MINIMAX-VALUE for E2, or else write “None” if there is no such allowed q .

Answer: For $p = \frac{1}{2}$, we have $E1 = 3$, and so we set $3 = 2q + (1 - q)6 = 6 - 4q$. Solving for q gives us $q = \frac{3}{4}$.

- ii. [3 points] Let $q = \frac{1}{4}$. Determine p such that the EXPECTED-MINIMAX-VALUE for E1 **is equal to** the EXPECTED-MINIMAX-VALUE for E2, or else write “None” if there is no such allowed p .

Answer: For $q = \frac{1}{4}$, we have $E2 = 2 \cdot \frac{1}{4} + 6 \cdot \frac{3}{4} = 5$. Setting $5 = p + (1 - p)5 = 5 - 4p$ and solving for p gives us $p = 0$.

- iii. [3 points] Let $q = 0$. Determine p such that the EXPECTED-MINIMAX-VALUE for E1 **is equal to** the EXPECTED-MINIMAX-VALUE for E2, or else write “None” if there is no such allowed p .

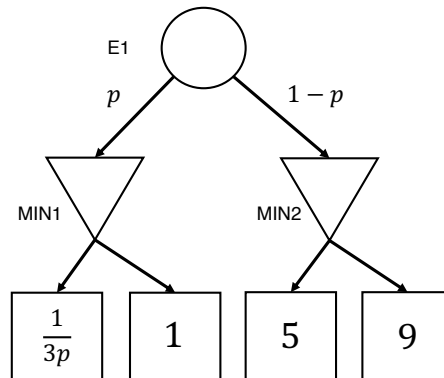
Answer: For $q = 0$, we have $E2 = 6$. Setting $6 = p + (1 - p)5 = 5 - 4p$ and solving for p , we have $p = -\frac{1}{4}$. However, this is outside the range $0 \leq p \leq 1$, and so the solution is “None”.

- (c) [2 points] Now consider the tree below, for which the probabilities, **as well as the utility of one of the terminal nodes**, are expressed in terms of p , where p **is only allowed to be** such that $0 \leq p \leq 1$.

For this tree, determine p such that the EXPECTED-MINIMAX-VALUE for E1 **is equal to** 3, or else write “None” if there is no such allowed p .

Answer: For $p \geq \frac{1}{3}$, the EXPECTED-MINIMAX-VALUE of MIN1 is $\min(1, \frac{1}{3p}) = \frac{1}{3p}$. Thus, for $p \geq \frac{1}{3}$, $E1 = \frac{1}{3} + (1 - p)5 = \frac{1}{3} + 5 - 5p$. Setting this equal to 3, we have $5p = \frac{7}{3}$, and solving for p gives us $p = \frac{7}{15}$.

Your Username:



3. [15 points] Constraint Satisfaction Problem

You come across a locked safe that requires a **5-digit passcode**. Taped to the top is a note containing information about the passcode. The information found in the note is as follows:

- (i) The password only consists of digits 1, 3, 5, and 7.
 - (ii) The third digit is unique.
 - (iii) The first digit is either 3 or 5.
 - (iv) The first and fourth digits are not the same.
 - (v) The first and fourth digits sum to 6.
 - (vi) The second and fifth digits sum to 8.
 - (vii) The fifth digit minus the third digit is less than or equal to 4.
- (a) [4 points] We will formulate the problem as a CSP in which there is one variable per digit, and we will denote these variables by: d_1 , d_2 , d_3 , d_4 , and d_5 . Please state the domains and constraints. Note that you need to show the domain **after applying the unary constraints**. Constraints (either implicit or explicit) should be specified formally and precisely.

Your Username:

Answer:

Domains: (1 points, -0.25 for each incorrect domain)

$$d_1 = \{3, 5\}$$

$$d_2 = \{1, 3, 5, 7\}$$

$$d_3 = \{1, 3, 5, 7\}$$

$$d_4 = \{1, 3, 5, 7\}$$

$$d_5 = \{1, 3, 5, 7\}$$

Constraints: (3 points: 1 point for first 4; 2 points for last 4 (-0.5 for each incorrect constraint among last 4))

$$d_3 \neq d_1$$

$$d_3 \neq d_2$$

$$d_3 \neq d_4$$

$$d_3 \neq d_5$$

$$d_1 \neq d_4$$

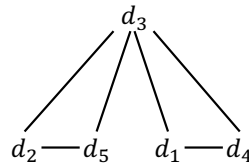
$$d_1 + d_4 = 6$$

$$d_2 + d_5 = 8$$

$$d_5 - d_3 \leq 4$$

- (b) [2 points] Draw the constraint graph of this CSP.

Answer: The constraint graph is shown as follows:



- (c) [1 points] Based on the constraint graph in (b), we may use cutset conditioning to solve this CSP. If we want to select only one node as the cutset, which node should we select?

Answer: We should select d_3 .

- (d) [6 points] You now notice an additional hint at the bottom of the note: “There are exactly four (distinct) accepted passcodes.” Find all four consistent solutions for this CSP. (Hint: use the cutset conditioning algorithm.)

Your Username:

Answer: We consider all four assignments of d_3 and compute residual CSP.
 $d_3 = 1$, residual domains: $d_1 = \{3, 5\}$, $d_2 = d_4 = \{3, 5, 7\}$, $d_5 = \{3, 5\}$. Arc consistency leads to failure, for $d_1 \rightarrow d_4$, $d_4 \rightarrow d_1$ arcs.
 $d_3 = 3$, residual domains: $d_1 = \{5\}$, $d_2 = d_4 = d_5 = \{1, 5, 7\}$. Arc consistency propagation yields $d_4 = \{1\}$, $d_2 = \{1, 7\}$, $d_5 = \{1, 7\}$. Resulting solutions: 5, 1, 3, 1, 7, and 5, 7, 3, 1, 1
 $d_3 = 5$, residual domains: $d_1 = \{3\}$, $d_2 = d_4 = d_5 = \{1, 3, 7\}$. Arc consistency propagation leads to failure, for $d_1 \rightarrow d_4$, $d_4 \rightarrow d_1$ arcs.
 $d_3 = 7$, residual domains: $d_1 = \{3, 5\}$, $d_2 = d_4 = d_5 = \{1, 3, 5\}$. Arc consistency propagation yields $d_1 = \{5\}$, $d_4 = \{1\}$, $d_2 = d_5 = \{3, 5\}$. Resulting solutions: 5, 3, 7, 1, 5, and 5, 5, 7, 1, 3

- (e) [2 points] Suppose you were informed of the following additional constraint: $(10d_4 + d_5)d_3 - (10d_1 + d_2) = 0$. Which of the solutions in (d) satisfies this additional constraint? (If there is no solution that satisfies this additional constraint, simply write “None”).

Answer: Of the four solutions, only 5, 1, 3, 1, 7 satisfies this additional constraint.

4. [15 points] First-Order Logic Inference

Consider the following statements:

- i. Everyone who loves Sam is happy
- ii. Sam is a dog
- iii. If Sam is an animal, then Carl loves Sam
- iv. A dog is an animal
- v. There is exactly one dog

In this problem, we would like to answer: **Is Carl happy?**

- (a) [4 points] Convert the sentences **(ii-v)** into first-order logic. As an example, sentence (i) can be translated into first-order logic as follows:

$$\forall x \text{ Loves}(x, \text{Sam}) \Rightarrow \text{Happy}(x)$$

You may use only the following predicates: $\text{Animal}(\cdot)$, $\text{Dog}(\cdot)$, $\text{Loves}(\cdot, \cdot)$, $\text{Happy}(\cdot)$.

Answer: One point each.

- ii. $\text{Dog}(\text{Sam})$
- iii. $\text{Animal}(\text{Sam}) \Rightarrow \text{Loves}(\text{Carl}, \text{Sam})$
- iv. $\forall x \text{ Dog}(x) \Rightarrow \text{Animal}(x)$
- v. $\exists x \text{ Dog}(x) \wedge \forall y \text{ Dog}(y) \Rightarrow (x = y)$

- (b) [1 points] To formulate this problem as a logical inference task, we need a “query”. What is the query, as expressed in first-order logic, for this problem?

Answer: $\text{Happy}(\text{Carl})$

Your Username:

- (c) [5 points] Convert your first-order logic sentences **(i-iv)** and the **query** into conjunctive normal form (CNF).

Answer: One point each.

- i. $\neg \text{Loves}(x, \text{Sam}) \vee \text{Happy}(x)$
- ii. $\text{Dog}(\text{Sam})$
- iii. $\neg \text{Animal}(\text{Sam}) \vee \text{Loves}(\text{Carl}, \text{Sam})$
- iv. $\neg \text{Dog}(x) \vee \text{Animal}(x)$

Query: $\text{Happy}(\text{Carl})$

- (d) [5 points] Prove whether the statements (i-iv) entail the query using resolution. (Remember to write out the substitutions.)

Answer:

First, let's add the negation of our conclusion to the KB: $\neg \text{Happy}(\text{Carl})$

One point each for steps (1) - (3), 2 points for step (4):

- (1) Resolve (ii) and (iv) with substitution $\{x/\text{Sam}\}$
- (2) Resolve with (iii)
- (3) Resolve with (i) with substitution $\{x/\text{Carl}\}$
- (4) Resolve with the negated goal to get an empty set.

5. [10 points] Short Answers (Explanation Needed)

- (a) [4 points] **True** or **False**: The propositional logic formula $(W \Rightarrow P) \wedge (P \vee W)$ is satisfiable. (Use a truth table to justify your answer.)

Answer: **True**.

W	P	$W \Rightarrow P$	$(P \vee W)$	$(W \Rightarrow P) \wedge (P \vee W)$
True	True	True	True	True
True	False	False	True	False
False	True	True	True	True
False	False	True	False	False

(Each row of the truth table is 1 point)

We can construct a truth table to show this. The explanation must be consistent with the answer.

- (b) [2 points] **True** or **False**: Both breadth-first search and depth-first search are cost-optimal if action costs are all the same.

Answer: **False**. (1 point) Breadth-first search is cost-optimal if action costs are all identical (and non-negative), but depth-first search may not be cost-optimal, even when all action costs are the same. (1 point)

Your Username:

- (c) [2 points] **True or False:** A utility-based agent maintains a binary distinction of “happy” vs. “unhappy”, but it cannot tell how happy it is.

Answer: **False.** (1 point) A goal-based agent maintains a binary distinction, whereas a utility-based agent has a measure of “happiness” (utility), which it then tries to maximize. (1 point)

- (d) [2 points] **True or False:** Uniform cost search is complete so long as all action costs are ≥ 0 .

Answer: **False.** (1 point) Uniform cost search is complete so long as all action costs are $\geq \epsilon > 0$ (1 point) (saying something to the effect of “strictly greater than 0” is also acceptable).

6. [15 points] Multiple Choice Questions (Explanation Not Needed)

For each of the following questions, only one choice is correct.

- (a) [3 points] Consider the function $f(x, y) = \frac{1}{2}x^2 + xy - 2y^2$. Let $(x_0, y_0) = (1, \frac{1}{2})$ be the starting guess state for an iterative algorithm for local search in a continuous state space. Which of the following would be the resulting state after **one** iteration of **gradient descent** with stepsize $\eta = \frac{1}{2}$?

- (A) $(1, \frac{1}{2})$
- (B) $(\frac{1}{4}, 0)$
- (C) $(\frac{1}{4}, 1)$
- (D) $(\frac{7}{4}, 0)$

Answer: **C).** $\nabla f(x, y) = (x + y, x - 4y)$.

$$\begin{aligned} (x_1, y_1) &= (x_0, y_0) - \eta \nabla f(x_0, y_0) \\ &= \left(1 - \frac{1}{2} \left(1 + \frac{1}{2}\right), \frac{1}{2} - \frac{1}{2} \left(1 - 4 \left(\frac{1}{2}\right)\right)\right) = \left(1 - \frac{3}{4}, \frac{1}{2} + \frac{1}{2}\right) = \left(\frac{1}{4}, 1\right) \end{aligned}$$

- (b) [3 points] Suppose we have a (minimax) tree with branching factor b and maximum depth m . Further suppose this tree is such that there exists a terminal node at depth d (for $d < m$). In this scenario, which of the following accurately characterizes the time and space complexities of the minimax search algorithm?
- (A) Time complexity: $O(b^d)$. Space complexity: $O(b^m)$.
 - (B) Time complexity: $O(bm)$. Space complexity: $O(b^m)$.
 - (C) Time complexity: $O(b^d)$. Space complexity: $O(bm)$.
 - (D) Time complexity: $O(b^m)$. Space complexity: $O(bm)$.

Answer: **D).** Explanation: The minimax algorithm performs an exhaustive depth-first exploration of the minimax tree.

- (c) [3 points] Consider a CSP, which by definition has a set of variables, a set of domains, and a set of constraints. Suppose we want to design a **new** CSP which has the **same** variables and domains as in the original CSP, but which now has, in addition to all of

Your Username:

the constraints as in the original CSP, **one more constraint**. Which of the following will **always** be a true statement about this **new** CSP?

- (A) The number of solutions of the new CSP will be strictly less than the number of solutions of the original CSP.
- (B) The number of solutions of the new CSP will be strictly greater than the number of solutions of the original CSP.
- (C) The number of solutions of the new CSP will be less than or equal to the number of solutions of the original CSP.
- (D) None of the above statements will always be true about this new CSP.

Answer: C). Any solution to the original CSP will either be unaffected by the new constraint (in which case the solution remains consistent), or else its assignment will become inconsistent when faced with this new constraint (in which case it is no longer a solution). Thus, a new constraint may (but does not have to) reduce the number of solutions for the new CSP, giving us the answer C).

- (d) [3 points] Consider the following set of clauses:

- i) $(X \vee C \vee \neg Y)$
- ii) $(\neg P \vee \neg C \vee X)$

Apply the resolution inference rule to derive a new clause. Which of the following clauses is the result of the application of the resolution inference rule?

- (A) $(X \vee \neg Y \vee \neg P \vee X)$
- (B) $(\neg P \vee \neg Y)$
- (C) $(C \vee \neg Y \vee \neg P \vee \neg C)$
- (D) $(C \vee \neg C)$

Answer: A). To find the result of the resolution, we look for complementary literals in different clauses. Here, we have complementary literals C in Clause i and $\neg C$ in Clause ii. Applying the resolution rule, we get the answer in A).

- (e) [3 points] Consider a local search problem in a discrete state space. Suppose our initial state is $(0,0)$, and the neighbors of our initial state are: $(-1,1)$, $(0,1)$, $(1,1)$. Further suppose the value at such states of the form (c_1, c_2) is computed using the following formula: $2c_1^2 + c_1c_2 - \frac{1}{2}c_2^2$. (For example, the value at $(0,0)$ is 0.) What would be the current state after one step of the **hill-climbing search algorithm**?

- (A) $(-1,1)$
- (B) $(0,1)$
- (C) $(1,1)$
- (D) $(0,0)$

Answer: C). The value at $(0,0)$ is 0. The value at $(-1,1)$ is $\frac{1}{2}$. The value at $(0,1)$ is $-\frac{1}{2}$. The value at $(1,1)$ is $\frac{5}{2}$, which is the greatest values among all of the neighbors' values (and it is also greater than the initial value), so that the answer is therefore C).